

C8000v IPsec lab — test evidence

Robot Framework run 2026-09-22_23-19-29 · every test runs against the live lab: the routers, the firewalls, the LAN hosts, Nautobot and the portal

66 / 74

tests passed

8

failed

0

skipped

12

suites

41 min

wall clock

Generated 2026-09-22T23:19:44.751762 · results in /home/dcantor/cat8000v-ipsec/results/2026-09-22_23-19-29 · 14 router configurations captured before and after the run, 483 lines of difference · log.html and report.html sit beside this file.

01 Management — 4 tests, 4 passed, 0 failed · 57s

Out-of-band management plane of the C8000v routers: reachability, SSH, RESTCONF, Mgmt-vrf, hardening.

Test	Result	Time	Evidence
Routers are reachable over the OOB network with SSH and RESTCONF	PASS	26s	3 packets transmitted, 3 received, 0% packet loss, time 2088ms rtt min/avg/max/mdev = 0.214/0.250/0.302/0.037 ms
Management interface lives in Mgmt-vrf with a default route to the host	PASS	7s	
License boot level provides the security feature set	PASS	4s	
AAA, SSH and VTY hardening from the NAC baseline	PASS	20s	

02 Underlay — 4 tests, 4 passed, 0 failed · 56s

WAN underlay: each spoke has a point-to-point /30 link to its headend's firewall, each headend one link to its firewall; the headend WAN address is reached through the firewall (static routes); LLDP adjacency; loopbacks.

Test	Result	Time	Evidence
WAN point-to-point interfaces are up with the modelled addresses	PASS	17s	
Each spoke reaches its headends' WAN addresses through the firewall, and nothing else	PASS	25s	Success rate is 100 percent (3/3), round-trip min/avg/max = 1/1/2 ms Success rate is 0 percent (0/2)
LLDP shows the firewall on each spoke's WAN port and on the headend's WAN port	PASS	10s	
Loopback0 carries the router-id and the LAN port the site LAN The site LAN /24 is the router's last port (.1 — the host's default gateway), no longer Loopback10.	PASS	4s	

03 Ipsec Vti — 5 tests, 5 passed, 0 failed · 102s

Point-to-point IPsec VTIs (tunnel mode ipsec ipv4) between each hub and each of its spokes, IKEv2 with pre-shared keys, traffic actually encrypted, no encryption of the OOB management plane.

Test	Result	Time	Evidence
Each tunnel is a static VTI sourced from the WAN link and protected by the IPsec profile	PASS	17s	

Tunnels are up/up at both ends and the tunnel subnet is reachable	PASS	16s	Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
IKEv2 SAs are READY with the modelled proposal and each spoke's chosen authentication (PSK or certificate) Auth sign / verify follow the spoke's choice (device ike_authentication, else the lab default): PSK, or RSA when it holds a certificate from the lab CA (suite 08 checks the certificates themselves). A headend runs one IKEv2 profile per method.	PASS	19s	
Traffic through the tunnels is encrypted with the modelled transform set	PASS	49s	Success rate is 100 percent (10/10), round-trip min/avg/max = 1/1/2 ms
Each hub holds exactly one IKEv2 session per spoke tunnel	PASS	2s	

04 Routing — 4 tests, 4 passed, 0 failed · 48s

eBGP over the VTIs: one AS per site, every hub peers with each of its spokes, spokes learn every other site via a hub (AS path through the hub), end-to-end reachability between site LANs.

Test	Result	Time	Evidence
Each hub has an Established eBGP session with each of its spokes over the tunnel	PASS	10s	Established, up for 00:49:36 Established, up for 1d05h
Every spoke peers with its hub(s) only and learns every other site	PASS	8s	
Spoke routes to other spokes go through a hub	PASS	6s	
Site LANs reach each other through the encrypted tunnels	PASS	23s	Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/5 ms

05 Nac Compliance — 1 tests, 1 passed, 0 failed · 48s

Configuration compliance: the routers match the Network-as-Code data model (no Terraform drift).

Test	Result	Time	Evidence
Terraform plan reports no drift from the NAC data model	PASS	48s	

06 Nautobot — 10 tests, 10 passed, 0 failed · 71s

The IPsec lab as modelled in the shared Nautobot: devices match the routers, p2p links are cabled, the VPN lives in the core VPN app (VPN, tunnels, hub/spoke endpoints, profile with Phase 1/2 policies), eBGP peerings match the live sessions, the rendered NAC device model matches the committed file, Golden Config is compliant.

Test	Result	Time	Evidence
Routers exist in Nautobot with role, serial, platform and management IP matching reality	PASS	14s	
WAN links are modelled as cables: spoke port to the headend's firewall, firewall to the headend, /30 addresses	PASS	3s	
The VPN is modelled in Nautobot's core VPN app: one VPN, one tunnel per hub/spoke pair, hub/spoke endpoints	PASS	12s	
Locations form the hierarchy lab site -> region -> branch, with site metadata on the branch	PASS	1s	

Every branch is a customer of ACME: a tenant per company on the router, its LAN host and its location, the address on the location, the ACME metadata and a design pattern that follows the tunnel count The headends belong to the provider (ACME Networks, tenant group Service provider); each branch to its customer company (tenant group Customers) with account, industry, tier and contract start as custom fields on the tenant. The branch location carries the street address; router and location carry the ACME design pattern, which is derived from how many tunnels (headends) the branch has: 1 = ACME-S	PASS	7s	
Every PSK spoke uses its own pre-shared key on all of its tunnels, each headend keys per PSK spoke and holds no key for a certificate spoke Each spoke chooses PSK or certificate (device ike_authentication, else the lab default): a keyring exists only on routers with a key-authenticated tunnel and lists exactly those peers (suite 08 proves the certificate side).	PASS	4s	
The IKEv2/IPsec suite comes from the VPN profile's Phase 1 / Phase 2 policies and matches the routers One VPN profile per authentication method in use (the lab default keeps the intent's names, the other one is suffixed); every tunnel references its spoke's profile; each router runs the profiles its tunnels use.	PASS	16s	
BGP model: one AS per site, eBGP peerings over the tunnel addresses matching the live sessions	PASS	4s	
NAC device model rendered from Nautobot matches the committed file	PASS	5s	
Golden Config: router backups are in Gitea and every compliance row is compliant	PASS	4s	

07 Firewalls — 8 tests, 8 passed, 0 failed · 108s

The VyOS firewall in front of each headend: modelled in Nautobot, configured from it, and actually filtering: only IKEv2, ESP and ICMP cross between the spokes and the headend; anything else is dropped.

Test	Result	Time	Evidence
Each headend has exactly one firewall, reachable over the OOB network with SSH	PASS	19s	3 packets transmitted, 3 received, 0% packet loss, time 2064ms rtt min/avg/max/mdev = 0.156/0.257/0.359/0.082 ms
Firewall interfaces carry the modelled addresses and face the right neighbours	PASS	5s	
The rendered firewall configuration matches what runs on the firewalls	PASS	26s	in sync fw-east: in sync fw-west: in sync
Only IKEv2, ESP and ICMP cross the firewall, IKEv2 and ESP only between the modelled WAN addresses; the tunnels actually carry ESP through it The address groups HEADEND-WAN / SPOKE-WAN hold exactly the WAN addresses Nautobot models on the far ends of the firewall's cables; the IKE / ESP rules reference them in both directions.	PASS	2s	

Firewall bandwidth is modelled in Nautobot and bounds the headend capacity together with the tunnel count Every firewall carries custom field <code>firewall_bandwidth_mbps</code> (from the intent); the VPN records the per-tunnel commitment; the portal's inventory reports both constraints per headend and the aggregate is the tighter of the two.	PASS	1s	
Headend CPU is collected live and joins the aggregate capacity The portal reads <code>`show platform resources`</code> on every headend: control-plane CPU against the platform's warning threshold, data-plane (QFP) CPU and DRAM. Aggregate utilisation is the tightest of tunnels, bandwidth and CPU.	PASS	8s	
Non-VPN traffic from a spoke to a headend is dropped by the firewall	PASS	32s	
The firewalls ship their firewall log to VictoriaLogs, and the portal reads the history from there The config context sets the remote syslog target (VictoriaLogs on the NMS, UDP 5514); <code>render_vyos</code> pushes it. The kernel's forward-filter lines arrive tagged with the firewall's hostname, the same lines <code>`show log firewall`</code> prints, so the portal's Firewalls tab (source=logs) and the Grafana drops panel / FirewallDropBurst alert see every firewall.	PASS	16s	

08 Pki — 7 tests, 6 passed, 1 failed · 684s

Certificate authentication, chosen per spoke (device `ike_authentication`, else the lab default): every router with a certificate-authenticated tunnel holds a certificate from the lab CA (`pki/ca.py`) in the modelled trustpoint, the CA is pinned by fingerprint, those tunnels authenticate with `rsa-sig`, and a router whose tunnels all use keys holds no trustpoint; Nautobot records what each router presents; the portal renews a certificate and switches a spoke between the two methods.

Test	Result	Time	Evidence
The lab CA exists and the routers with certificate tunnels trust exactly it, pinned by fingerprint	PASS	11s	
Every router with a certificate tunnel holds a certificate issued by the lab CA to its own name, valid and not close to expiry; key-only routers hold none	PASS	6s	
Certificate tunnels authenticate with <code>rsa-sig</code> against the trustpoint; a key stays only where a spoke chose one	PASS	12s	
The NaC data carries a key only for the spokes that chose one, and the rendered model matches Nautobot	PASS	0s	
Nautobot records the certificate every router presents	PASS	2s	
The portal renews a spoke's certificate: new serial on the router, in the CA index and in Nautobot; every tunnel back with RSA A real renewal of the first spoke through the portal's <code>renew</code> run (operator): the router's tunnels re-authenticate with the new certificate, the old serial is retired into the index's history.	PASS	52s	

The portal switches a spoke between certificate and pre-shared key and back: profiles, keyring, trustpoint, SAs and Nautobot follow A certificate spoke goes to its pre-shared key through the portal's auth run (its trustpoint retired, its headends keyring for it on their PSK profile, SAs on PSK) and back again (enrolled anew, keyring gone, SAs on RSA) — two runs, about ten minutes. Keyword 'Run Finished' failed after retrying for 10 minutes. The last error was: run 2026-09-22_23-29-27-b6ce still running	FAIL	601s	
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09 Hosts — 4 tests, 2 passed, 2 failed · 37s

The LAN hosts: one small Alpine VM behind every router (its eth1 = .2 of the site LAN, the router's LAN port .1 as the gateway), modelled in Nautobot and cabled to the router; every host reaches every other host — branch to branch through a headend, branch to headend, headend to headend — over the IPsec tunnels (full ping mesh), and the path shows the tunnel.

Test	Result	Time	Evidence
Every router has a LAN host, reachable over the OOB network, addressed on the site LAN with the router as its gateway	PASS	18s	3 packets transmitted, 3 received, 0% packet loss, time 2086ms rtt min/avg/max/mdev = 0.179/0.329/0.590/0.185 ms
The hosts are modelled in Nautobot: device at the router's site, eth1 addressed and cabled to the router's LAN port	PASS	0s	
Every host reaches every other host, its own router and the internet: the full ping mesh over the tunnels Every ordered pair (42 for 7 hosts): branch to its headends, branch to branch through a shared headend, headend to headend through a spoke homed on both (headends do not peer with each other) — plus, per host, its own router (the LAN gateway) and the internet (1.1.1.1 through its headend's breakout): 56 checks. [('host-central', 'host-spoke1'), ('host-east', 'host-spoke1'), ('host-west', 'host-spoke1'), ('host-spoke1', 'host-acquisition'), ('host-spoke1', 'host-central'), ('host-spoke1', 'host-east'), ('host-spoke1', 'host-west'), ('host-spoke1', 'host-spoke2'), ('host-spoke1', 'host-spoke3'), ('host-spoke1', 'host-spoke4'), ('host-spoke1', 'internet'), ('host-spoke2', 'host-spoke1'), ('host-spoke3', 'host-spoke1'), ('host-spoke4', 'host-spoke1')] of 72 checks failed: [('host-central', 'host-spoke1'), ('host-east', 'host-spoke1'), ('host-west', 'host-spoke1'), ('host-spoke1', 'host-acquisition'), ('h	FAIL	19s	
The path between two branch hosts goes through the branch routers and a headend's tunnels no headend tunnel address in the path: ['172.17.1.1', '172.17.4.1']	FAIL	0s	

10 Portal — 11 tests, 6 passed, 5 failed · 1129s

The portal itself: runs queue instead of colliding (one executes at a time, a queued one can be cancelled), every router has a page that gathers its tunnels, authentication, firewall rules, host and runs, and single sign-on (OpenID Connect against the lab's Gitea) is offered and starts a proper authorization-code + PKCE flow.

Test	Result	Time	Evidence
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Runs queue behind the one executing and a queued run can be cancelled; the executing one finishes <code>2 != 1</code>	FAIL	0s	
Every router has a page: identity, authentication, its tunnels with live state, firewall rules touching it, its host and its runs <code>central-headend: central-headend-spoke1 is {'like_status': 'none', 'ike_active_s': None, 'line_protocol': 'down', 'admin': 'administratively down', 'bgp_state': 'Idle', 'bgp_prefixes': None, 'bgp_updown': '00:03:13', 'encaps': None, 'decaps': None, 'send_errors': None, 'recv_errors': None, 'in_rate_bps': 0, 'out_rate_bps': 0, 'last_input': '00:03:18', 'last_output': '00:03:18', 'health': 'down'}: down != up</code>	FAIL	67s	
The Branches list names every router with its health, method, certificate and host, and each row opens the router's page <code>psk != certificate</code>	FAIL	2s	
Every router's page shows its configuration: the live running config (keys redacted for viewers), Nautobot's intended config and compliance	PASS	25s	
A router's page shows its configuration history from Gitea, the diff of a backup commit, and live show snippets	PASS	3s	
The Compliance page reports Nautobot's Golden Config verdict for every router and feature, and a golden run refreshes it <code>Keyword 'Run Finished' failed after retrying for 8 minutes. The last error was: run 2026-09-22_23-41-43-58bb still queued</code>	FAIL	481s	
The API describes every field of the intent: a round trip through it loses nothing Several run modes save the intent the caller posted, and the portal serialises it through the REST schema — a field the schema does not know is silently dropped and the model loses it (that is how the DCI's NAT and DNS settings disappeared once). Whatever is in lab-intent.json must survive GET /api/intent.	PASS	0s	
Provisioning a new router can register it in lab.conf: every array the pipeline rewrites is found and reads back The first step of a spoke / headend run writes the new router into lab.conf's arrays. They are wrapped over several lines (every router has a LAN host), so the rewriter has to span lines — this is a dry run of that rewrite on the real lab.conf, read back with bash; lab.conf itself is untouched.	PASS	0s	

Drift on a router is detected by the Golden Config run, remediated from the portal with Nautobot's remediation lines, and recorded in the drift history An extra static route is configured on a spoke by hand. The golden run marks its Static routes feature non-compliant with the `no ...` remediation line; a `remediate` run pushes that line and saves, the next verdict is compliant again, the drift history shows the drift and its repair, and Prometheus (via /metrics) saw the router non-compliant in between. 'False' should be true.	FAIL	253s	<code>compliant': True, 'missing': '', 'extra': '', 'remediation': '', 'diff': '---\nrunning\n+++ intended\n@@ -2,18 +2,18 @@\ncompliant': 19, 'total': 19, 'drifted': []}, {'at': '2026-09-22 22:19:40', 'ok': True, 'compliant': 19, 'total': 19, 'drifted':</code>
Re-apply from the model restores a modelled attribute that was changed by hand, through a Terraform apply targeted at that router Loopback0's description on a spoke is changed by hand (Terraform manages it). A `reapply` run renders the NaC data, plans and applies only that router's resources, and ends with Golden Config compliant; the description is back.	PASS	297s	
Single sign-on is offered and starts an authorization-code flow with PKCE at the provider	PASS	0s	

11 Internet — 4 tests, 4 passed, 0 failed · 53s

Internet breakout: every firewall's uplink sits on the libvirt NAT network and masquerades the site LANs (LAN -> uplink only, replies routed back to the headend), every headend holds a default via its firewall and originates one to its spokes, every spoke prefers the nearest headend's default (local preference from the intent's regions), and every LAN host reaches the internet through its preferred headend — the path shows it and the firewall logs it.

Test	Result	Time	Evidence
Every firewall has its uplink on the NAT network with an address, masquerades the site LANs and routes them back to its headend	PASS	18s	2 packets transmitted, 2 received, 0% packet loss, time 1002ms rtt min/avg/max/mdev = 15.555/16.646/17.737/1.091 ms
Every headend holds a static default via its firewall and originates a default route to each of its spokes	PASS	6s	
Every spoke prefers the nearest headend's default route (local preference by region) and keeps the others as backups	PASS	11s	
Every LAN host reaches the internet through its router's preferred headend, and the firewall there logs the flow A branch host breaks out through its nearest headend. A host behind the DCI chain has no breakout preference of its own (its default route comes from the headend over eBGP) — suite 12 checks that path.	PASS	18s	

12 Dci — 12 tests, 12 passed, 0 failed · 81s

The DCI chain off west-headend: ACME's data-centre interconnect (DCI) and the acquired company's edge router (ACME-acquisition) behind it. They carry no IPsec — west-headend <-> DCI <-> ACME-acquisition are direct /30 links with plain eBGP — and the acquisition originates its own prefixes (its site LAN and a service prefix on a loopback), which must be routable from every branch over the VPN. Modelled in Nautobot like everything else (devices, cables, prefixes, BGP peerings, tenants) and rendered by the same NaC pipeline and Golden Config template.

Test	Result	Time	Evidence
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The DCI chain is modelled in Nautobot: roles, tenants, cabled links, prefixes and the acquisition's own service prefix ACME owns the DCI (tenant ACME Networks, role vpn-dci); the acquired company is a customer tenant of its own on a partner-edge router. The two links are cables between router ports (no firewall in between) and their /30s are modelled.	PASS	1s	
The interconnect links are up with the modelled addresses, no IPsec anywhere on them, and the neighbour answers Plain routed links: the addresses come from the model, the far end pings, and neither router holds a tunnel, a crypto profile or a keyring — the encryption stops at the headends.	PASS	7s	Success rate is 100 percent (3/3), round-trip min/avg/max = 2/4/5 ms
eBGP is Established on every hop of the chain with the modelled AS numbers west-headend <-> DCI <-> ACME-acquisition: one session per link, over the link addresses (not loopbacks), each with the AS from the model — the same nautobot-bgp-models peerings the tunnels use, just over a direct link.	PASS	3s	
The acquisition originates its prefixes and every branch learns them through its headend The site LAN and the service prefix on the loopback are `network`ed by ACME-acquisition, cross the DCI into west-headend's AS and are re-advertised over the tunnels: every branch has them via one of its headends and can reach them.	PASS	6s	
A branch reaches the acquisition's LAN and its service prefix, and the acquisition reaches every branch End to end over IPsec + the DCI: a branch pings both acquisition prefixes from its own LAN address, and the acquisition's host reaches every branch host (the ping mesh of suite 09 covers the reverse direction as well).	PASS	6s	Success rate is 100 percent (3/3), round-trip min/avg/max = 4/4/5 ms
The host behind the acquisition is in the mesh and breaks out to the internet through west-headend Its default route comes from the headend over eBGP (it has no breakout preference of its own), so the path runs acquisition -> DCI -> west-headend -> the firewall that NATs.	PASS	9s	
Golden Config renders the DCI chain from the same template and reports it compliant The two routers are in the Golden Config scope like every other Catalyst 8000v: the template renders them without a crypto section (they have no VPN profile) and every compliance feature matches.	PASS	12s	compliant': True, 'missing': '', 'extra': '', 'remediation': '', 'diff': '--- running\n+++ intended\n@@ -8,6 +8,6 @@\naddress
The overlapping prefix exists on both sides of the DCI and neither side learns the other's copy The acquired company runs a server on a prefix ACME already uses (a branch LAN). The DCI filters that prefix in both directions — each side keeps exactly one path for it, its own — and advertises the translated range instead.	PASS	4s	
The DCI translates in both directions: each side reaches the other through the range it was given ACME reaches the acquisition's server at the inside-global address and the acquisition reaches ACME's host at the outside-local one; the translations show up in the NAT table with both halves.	PASS	7s	Success rate is 100 percent (3/3), round-trip min/avg/max = 3/3/4 ms

DNS is fixed up on the way through the NAT: a name that resolves to an overlapping address comes back translated ACME's headend answers for its own zone; the acquisition resolves a name whose A record is an address that exists on both sides, and the DCI's NAT rewrites the answer to the range the acquisition can actually reach.	PASS	2s	Success rate is 100 percent (3/3), round-trip min/avg/max = 5/5/6 ms
NAT scales: a thousand overlapping prefixes are translated in both directions with no drops The scale set puts a thousand prefixes on both sides at once, each with its own pair of static translations on the DCI (plus the two single ones), while BGP only carries the three aggregates — the scale lands on the NAT, not the routing table. A branch host sweeps every translated prefix and the acquisition sweeps a sample back.	PASS	7s	Total active translations: 3020 (2002 static, 1018 dynamic; 2 extended) Outside interfaces: GigabitEthernet2 Inside interfaces: GigabitEthernet3 Hits: reached 1000/1000 Success rate is 100 percent (2/2), round-trip min/avg/max = 2/2/2 ms
DNS fix-up scales and works from either side: a host on one side resolves the other side's zone through the NAT Each side answers for its own zone with its own copies of the overlapping prefixes — a thousand records each. A branch host on the ACME side resolves every name in the acquired company's zone (its resolver is the server's translated address) and gets addresses it can actually reach; the acquisition resolves ACME's zone the same way. Both are rewritten by the DCI in flight.	PASS	16s	resolved+reached 1000/1000 Success rate is 100 percent (2/2), round-trip min/avg/max = 3/3/4 ms